

Galaxy B Modes

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I Introduction and Motivation

TODO

II Bias, Effective Field Theory (EFT), Intrinsic Alignment

II.1 Halo Bias and EFT

In this section we introduce the much more established field of galaxy bias and an EFT of galaxy bias. This will directly translate to the EFT used in section II.3. We consider a Λ CDM universe with a perturbed FLRW metric unless stated otherwise.

II.1.1 What is bias

Say we look at the universe at redshift z . Given a perturbation to the matter density $\delta(\mathbf{x}, z)$ we would like to know the perturbation to the density of some other tracer. In this section we will consider the relative number overdensity of dark matter halos, δ_h , but the results are applicable to any tracer in general, notably to galaxy number density. We then model the relation between these two as

$$\delta_h(\mathbf{x}, z) = \sum_O b_O(z) O(\delta)(\mathbf{x}, z), \quad (\text{II.1})$$

where we sum over some set of operators $\{O\}$ that map δ to a new function of position and time and the b_O are the BPs. Crucially, they only depend on z and not on $\mathbf{x}$. 4 questions arise naturally from this model:

1. What is the motivation to use this model? Why is this expected to be accurate up to a (reasonably small) scale cutoff with the right choice of operators and values of BPs?
2. How can we choose a set of operators that is "complete"? I.e. that the model is correct in the sense of the previous question.
3. How do the bias parameters evolve over time?
4. How do we determine bias parameters? Both theoretically and empirically given a dataset from a real life survey or a simulation.

The rest of this section will be devoted to summarizing the answers to these questions.

II.1.2 Relating bias parameters from different times

We consider a simple case where a population of galaxies instantly formed at some time τ_* . This is sufficient because the evolution equations we obtain will then work as a “Green’s function” for the evolution equations of bias parameters in general. We simply integrate over the time evolved bias parameters weighted by the rate of galaxy formation at the integrand time.

We assume that whatever tracer we consider moves with the fluid. This immediately gives the continuity equation:

$$\frac{D}{D\tau} \delta_g = -\theta(1 + \delta_g), \quad (\text{II.2})$$

where

$$\frac{D}{D\tau} = \frac{\partial}{\partial \tau} + v^i \frac{\partial}{\partial x^i} \quad (\text{II.3})$$

is the convective time derivative, v^i is the peculiar velocity of the cosmic matter fluid, and $\theta = \partial_i v^i$ is the velocity divergence. To evolve δ_g we also need evolution equations for δ and v^i . They are given by the continuity equation (again)

$$\frac{D}{D\tau}\delta = -\theta(1 + \delta), \quad (\text{II.4})$$

and the Euler equation,

$$\frac{D}{D\tau}\theta = -\mathcal{H}\theta - (\partial^i v^j)^2 - \frac{3}{2}\Omega_m \mathcal{H}^2 \delta. \quad (\text{II.5})$$

We proceed by dividing equations II.4 and II.2 by $1 + \delta$ and $1 + \delta_g$, respectively to get

$$\frac{1}{1 + \delta_g} \frac{D}{D\tau} \delta_g = \frac{1}{1 + \delta} \frac{D}{D\tau} \delta. \quad (\text{II.6})$$

This can be solved by switching to Lagrangian coordinates to get

$$\ln(1 + \delta_g(\mathbf{x}(\tau), \tau)) = \ln(1 + \delta(\mathbf{x}(\tau), \tau)) + \ln\left(\frac{1 + \delta_g(\mathbf{x}(\tau), \tau)}{1 + \delta(\mathbf{x}(\tau), \tau)}\right), \quad \tau > \tau_*, \quad (\text{II.7})$$

which can be written more simply as

$$1 + \delta_g|_{\tau} = \frac{1 + \delta|_{\tau}}{1 + \delta|_{\tau_*}} (1 + \delta_g|_{\tau_*}). \quad (\text{II.8})$$

This equation can be made usefull by expanding order by order. Up to second order we get

$$1 + \delta_g^{(1)}(\mathbf{x}, \tau) + \delta_g^{(2)}(\mathbf{x}, \tau) = 1 + \delta^{(1)} - \delta_*^{(1)} + \delta_{g*}^{(1)} + \delta^{(2)} - \delta(2)_* + \delta_{g*}^{(2)} + [\delta_*^{(1)}]^2 - \delta^{(1)}\delta_*^{(1)} + \left[\delta^{(1)} - \delta_*^{(1)}\right] \delta_{g*}^{(1)}. \quad (\text{II.9})$$

Now, the difference between $\mathbf{x}$ and $\mathbf{x}_*$ is first order in perturbations as well. We can Taylor expand around $\mathbf{x}$ and use that the displacement, $\mathbf{x} - \mathbf{x}_*$ evolves according to the growth factor D to get

$$\delta_g^{(1)}(x, \tau) = \left(1 + \frac{D_*}{D} [b_1^* - 1]\right) \delta^{(1)}(x, \tau) + \varepsilon^* \quad (\text{II.10})$$

$$\delta_g^{(2)}(x, \tau) = \left\{1 + [b_1^* - 1] \left(\frac{D_*}{D}\right)^2\right\} \delta^{(2)} + \left\{\frac{D_*}{D} [b_1^* - 1] - \left(\frac{D_*}{D}\right)^2 [b_1^* - 1] + \frac{1}{2} b_2^* \left(\frac{D_*}{D}\right)^2\right\} [\delta^{(1)}]^2 \quad (\text{II.11})$$

$$+ b_K^{*2} \left(\frac{D_*}{D}\right)^2 [K_{ij}^{(1)}]^2 + \left(\frac{D_*}{D} - 1\right) \frac{D_*}{D} [b_1^* - 1] s_{(1)}^i \partial_i \delta^{(1)} - \left(\frac{D_*}{D} - 1\right) \varepsilon^* \delta^{(1)} \quad (\text{II.12})$$

$$+ \left(\frac{D_*}{D} - 1\right) s_{(1)}^i \partial_i \varepsilon^*. \quad (\text{II.13})$$

Here, ε is a stochastic parameter added in and $s^i = x^i - x_*^i$ is the displacement. One can then extract the bias parameters at current time (Eularian time) by comparing to the bias expansion at Eularian time and matching terms to get

$$b_1^E = 1 + \frac{D_*}{D} (b_1^* - 1) \quad (\text{II.14})$$

$$b_2^E = b_2^* \left(\frac{D_*}{D}\right)^2 + \frac{8}{21} \left(1 - \frac{D_*}{D}\right) (b_1^E - 1) \quad (\text{II.15})$$

$$b_{K^2}^E = b_{K^2}^* \left(\frac{D_*}{D}\right)^2 - \frac{2}{7} \left(1 - \frac{D_*}{D}\right) (b_1^E - 1) \quad (\text{II.16})$$

Clearly this can be easily generalized to any order.

II.1.3 EFT of bias parameters

Main ref: [1] We would like to find a "basis" of bias parameters up to a given order in perturbation theory. We will assume that (1) gravitation is described by general relativity, (2) that we can neglect the impact of massive neutrinos and dark energy perturbations, (3) that initial conditions are Gaussian and adiabatic.

We will assume that the only quantities relevant for the formation of galaxies are local on some scale R_* (which is much larger than the size of a galaxy / whatever tracer we are considering). According to the equivalence principle, the only effects that can be measured by a freely falling observer (i.e. forming galaxy) is the second derivatives of the gravitational potential, $\partial_i \partial_j \Phi$. Higher derivatives will be higher order contributions, because each derivative on a scale of R_* contributes a factor of k/R_* , which for large enough scales will be much smaller than one.

Now, the galaxy density, δ_g will in a very general sense depend on the history of $\partial_i \partial_j \Phi$ along the fluid trajectory. In other words:

$$\partial_g(\mathbf{x}, \tau) \supset \int^\tau d\tau' f_O(\tau, \tau') O(\mathbf{x}_{fl}(\tau'), \tau'), \quad (\text{II.17})$$

where f_O is some kernel that we don't need to bother defining. We can then Taylor expand to get

$$\partial_g(\mathbf{x}, \tau) \supset \left[\int^\tau d\tau' f_O(\tau, \tau') \right] O(\mathbf{x}_{fl}(\tau), \tau) + \left[\int^\tau d\tau' (\tau' - \tau) f_O(\tau, \tau') \right] \frac{D}{D\tau} O(\mathbf{x}_{fl}(\tau), \tau) + \dots, \quad (\text{II.18})$$

where $D/D\tau$ is the convective derivative along the fluid flow. In order to define a basis we thus also have to include time derivatives $D/D\tau$ of the second derivatives of the potential. It now seems like we would need an infinite set of operators to form a basis. This can be avoided by noticing at finite order in perturbation theory, only a finite amount of these operators are linearly independent of one another.

In order to show the above statement, we will first switch to Lagrangian coordinates $\mathbf{q} = \mathbf{x}_{fl}(\tau = 0)$ where the convective time derivative simplifies to $\partial/\partial\tau$. We also assume that the n -th order growth factor is given by the linear growth factor to the n -th power. This is only strictly valid for an EdS (flat matter dominated universe) but is still very accurate for other cosmologies such as Λ CDM. Consider some operator in the basis, $O_L(\mathbf{q}, \tau)$ which has contributions at various orders, $O_L^i(\mathbf{q}, \tau)$, then

$$\left(\frac{D}{D\tau} \right)^n O_L(\mathbf{q}, \tau) |^{(n)} = \left(\frac{\partial}{\partial\tau} \right)^n O_L(\mathbf{q}, \tau) |^{(n)} = \sum_{i=1}^n \left(\frac{d^n}{d\tau^n} D^i(\tau) \right) O_L^i(\mathbf{q}, \tau_0). \quad (\text{II.19})$$

Next, although higher order contributions start being linearly dependent on the lower order contributions after some order, as shown above, it is worth noting that these higher order contributions are in general no longer local in $\partial_i \partial_j \phi$. This is immediately obvious if we notice that the convective time derivative contains a spatial derivative that generates higher order (k/R_*) terms. It turns out that, due to symmetry requirements, we only need to consider $\partial_i \partial_j / \nabla^2$ acting on powers of $\partial_k \partial_l \Phi$ in order to get all the new terms created by the extra spatial derivatives.

Constructing a *local* Eulerian basis out of $\partial_i \partial_j \Phi(\mathbf{x}, \tau)$ and its convective time derivatives can be done explicitly as first shown in [2]. First define

$$\Pi_{ij}^{[1]}(\mathbf{x}, \tau) = \frac{2}{3\Omega_m \mathcal{H}^2} \partial_i \partial_j \Phi(\mathbf{x}, \tau) = K_{ij}(\mathbf{x}, \tau) + \frac{1}{3} \delta_{ij} \delta(\mathbf{x}, \tau) \quad (\text{II.20})$$

and

$$\Pi_{ij}^{[n]} = \frac{1}{(n-1)!} \left[\frac{1}{\mathcal{H}f} \frac{D}{D\tau} \Pi_{ij}^{[n-1]} - (n-1) \Pi_{ij}^{[n-1]} \right], \quad (\text{II.21})$$

as given by ref. [2]. Then take into account that $\text{tr} \Pi^{[n]}$ can be written in terms of lower order operators through the Eulerian fluid equations.

To get all terms at order n , we then write products of $\Pi^{[k]}$ where $k < n$ and the total order adds up to n or less and take traces of the different possible combinations of each product.

Thusfar we have only considered an expansion local in the second derivatives of the gravitational potential. Higher order derivatives can and should also be considered. Assume that all relevant non-local effects are limited to some scale, the "scale of nonlocality" R_* . Following the reasoning in [3], nonlocal contributions to the tracer number density at $\mathbf{x}$ will come from terms at $\mathbf{x} + \mathbf{y}$, where $y \sim R_*$. Take, for example, $\delta(\mathbf{x} + \mathbf{y})$. Through a Taylor expansion at $\mathbf{x}$ we thus find that this contribution can be written as

$$b_{\delta(\mathbf{x}+\mathbf{y})} \delta(\mathbf{x} + \mathbf{y}) = b_{\delta(\mathbf{x}+\mathbf{y})} (\delta(\mathbf{x}) + \mathbf{y} \cdot \vec{\nabla} \delta(\mathbf{x}) + y^2 \nabla^2 \delta(\mathbf{x}) + \dots).$$

Terms with an odd order of derivatives will not contribute due to isotropy. On the other hand, $y^2 \nabla^2 \sim R_*^2 k^2$ in Fourier space. We thus see that for scales much larger than the non-locality scale (the regime in which we expect biasing to work anyway), higher derivative terms are suppressed through the factor $R_*^{2n} k^{2n} \ll 1$, where n is the order of the derivative. Note that we assume that the BP's themselves are not much larger than 1, and if we find that that is not the case then there's a sign that our perturbative expansion has stopped working for the scales considered. When aiming to find a full basis for our expansion up to some order, we thus need to consider the tracer in question to know up to which order in derivatives and perturbation theory to work.

II.2 Spin-2 Fields and Their Projections

It is well known how to construct powerspectra from scalar quantities. Constructing them for a spin-2 field requires more care. In particular, we would like to find a way to define powerspectra that is independent of the orientation of the observer.

We model galaxies in 3D space as ellipsoids where we do not care about the overall size. This means they can be described according to the 3×3 symmetric trace-free tensor Q_{ij} given by

$$Q_{ij} = \text{tf} \left(\frac{\int d^3x \rho(\mathbf{x}) x^i x^j}{\int d^3x \rho(\mathbf{x})} \right), \quad (\text{II.22})$$

where we assume for simplicity that the galaxy's center of mass is at $x^i = 0$ and $\text{tf}(\cdot)$ means taking the trace free part. The trace corresponds to the overall size of the galaxy and is not of interest to us. This corresponds roughly to g_{ij} from [4]. Q_{ij} at some point can transform under $\text{SO}(3)$. Specifically, it transforms under the spin-2 representation, which can immediately be seen because Q_{ij} is a 5 dimensional vector space and there is only one irreducible representation of $\text{SO}(3)$ of dimension $2\ell + 1$, with in this case $\ell = 2$. This is the classification theorem. Alternatively, it is clear that under a $\text{SO}(3)$ rotation given by R , we get $Q_{ij} \rightarrow R_{ia} R_{jb} Q_{ab}$. One can compare this to the spin-2 representation of $\text{SO}(3)$ to see that they are the same.

We observe shapes as projected onto the sky. If we take $\mathbf{n}$ as our line of sight with associated projec-

tion matrix P^{ij} , then

$$\gamma_{ij} := \text{tf}_P \left(P^{ik} P^{jl} g_{kl}(\mathbf{x}, z) \right) \quad (11.23)$$

$$= \frac{1}{2} \left(P^{ik} P^{jl} + P^{il} P^{jk} - P^{ij} P^{kl} \right) g_{kl}(\mathbf{x}, z) \quad (11.24)$$

$$=: P^{ijkl} g_{kl}(\mathbf{x}, z). \quad (11.25)$$

In the last line we simply define P^{ijkl} as the combination of P^{ij} from the line above. Going from line 1 to line 2 follows by using that in the projected space, the metric is P^{ij} instead of δ^{ij} and the trace of an arbitrary tensor M_{ij} thus becomes $P^{ij} M_{ij}$.

γ_{ij} is again symmetric, trace-free, and transforms like a spin-2 representation, now of $\text{SO}(2)$.

11.2.1 Spherical tensor decomposition

Going back to the non-projected Q_{ij} , it's clear that this tensor is defined at each point in space, $\mathbf{x}$. Now look at the Fourier transform, $Q_{ij}(\mathbf{k})$. For a mode in a given direction, $\hat{\mathbf{k}}$, we can decompose $Q_{ij}(\mathbf{k})$ through the spherical tensors $(Y_2^{(m)})(\hat{\mathbf{k}})_{ij}$, with $m = -2, -1, 0, 1, 2$ denoting the helicity. We get

$$(Y_2^{(0)})(\hat{\mathbf{k}})_{ij} = \sqrt{3/2}(\hat{\mathbf{k}}_i \hat{\mathbf{k}}_j - \frac{1}{3} \delta_{ij}) \quad (11.26)$$

$$(Y_2^{(\pm 1)})(\hat{\mathbf{k}})_{ij} = \sqrt{1/2}(\hat{\mathbf{k}}_i \mathbf{e}_j^\pm + \hat{\mathbf{k}}_j \mathbf{e}_i^\pm) \quad (11.27)$$

$$(Y_2^{(\pm 1)})(\hat{\mathbf{k}})_{ij} = \mathbf{e}_i^\pm \mathbf{e}_j^\pm \quad (11.28)$$

with $e^\pm = \dots$ and $i, j = 1, 2, 3$. The unit vectors e_1 and e_2 are defined by ... For a rotation around $\hat{\mathbf{k}}$ by an angle ϕ the spherical tensors transform as

$$(Y_2^{(m)})(\hat{\mathbf{k}})_{ij} \rightarrow e^{im\phi} (Y_2^{(m)})(\hat{\mathbf{k}})_{ij}.$$

These spherical tensors have a number of other usefull properties which we will not get into right now.

11.2.2 Defining Power Spectra

We can now define the powerspectrum as a $3 \times 3 \times 3 \times 3$ tensor given by

$$\langle Q_{ij}(\mathbf{k}) Q_{ij}(\mathbf{k}') \rangle = (2\pi)^3 \delta^D(\mathbf{k} + \mathbf{k}') P_{ijkl}^{QQ}(\mathbf{k}). \quad (11.29)$$

This definition is the most obvious, however because Q_{ij} transforms as a tensor under rotations, it leaves $P_{ijkl}^{QQ}(\mathbf{k})$ dependent on the direction of $\mathbf{k}$. We can instead work with the decomposition of Q_{ij} in the spherical tensors:

$$Q_{ij}(\mathbf{k}) = \frac{1}{3} \delta_{ij} Q_0^0(\mathbf{k}) + \sum_{m=-2}^2 Q_2^m(\mathbf{k}) Y_{ij}^{(m)}(\hat{\mathbf{k}}), \quad (11.30)$$

where we now allow Q_{ij} to also have a trace / scalar component Q_0^0 . We can then define the following powerspectra:

$$\langle S_l^{(m)}(\mathbf{k}) S_{l'}^{(m')}(\mathbf{k}') \rangle = (2\pi)^3 \delta_{mm'} \delta^D(\mathbf{k} + \mathbf{k}') P_{ll'}^{(m)}(k). \quad (11.31)$$

When the powerspectra are defined in this way, they are scalars. There are 7 independent spectra, however we also have by the properties of the spherical tensors that

$$P_{ll'}^{(m)}(k)^* = P_{ll'}^{(-m)}(k). \quad (11.32)$$

If we also enforce invariance under parity transformations, then we find that the powerspectra are real and thus that

$$P_{ll'}^{(m)}(k) = P_{ll'}^{(-m)}(k). \quad (II.33)$$

This reduces the amount of independent spectra to 5.

II.2.3 Noise power spectra

In accordance with the previous subsection, there are 3 power spectra due to stochastic / noise terms in the EFT of galaxy IA. They come from combinations of scalar / galaxy size noise and spin-2 / symmetric traceless tensor / galaxy shear noise.

$$\langle \epsilon(\mathbf{k}) \epsilon(\mathbf{k}') \rangle' = P_\epsilon^s(k), \quad (II.34)$$

$$\langle \epsilon_{ij}(\mathbf{k}) \epsilon(\mathbf{k}') \rangle' = (\mathbf{k}_i \mathbf{k}_j - \frac{1}{3} \delta_{ij} k^2) P_\epsilon^{gs}(k), \quad (II.35)$$

$$\langle \epsilon_{ij}(\mathbf{k}) \epsilon_{kl}(\mathbf{k}') \rangle' = (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} - \frac{2}{3} \delta_{ij} \delta_{kl}) P_\epsilon^g(k). \quad (II.36)$$

In all three cases $P_\epsilon^{\dots}(k) = c^{\dots} + \mathcal{O}(R^2 k^2)$. The forms of the spectra are again constructed to satisfy transformation properties, symmetry, and tracelessness. For the scalar-scalar and tensor-tensor spectra the above equations correspond to white noise with some corrections terms that only become significant if the scales we are looking at ($1/k$) become comparable to the scale of nonlocality of galaxy formation (R). For the tensor-scalar correlation there is no ‘white noise’ (or term independent of k). To see why, consider what would happen if $P_\epsilon^{gs}(k)$ would have a k^{-2} term.

$$\langle \epsilon_{ij}(\mathbf{k}) \epsilon(\mathbf{k}') \rangle' \supset \left(\frac{\mathbf{k}_i \mathbf{k}_j}{k^2} - \frac{1}{3} \delta_{ij} \right) \xrightarrow{\mathcal{F}^{-1}} \propto (\partial_i \partial_j \nabla^{-2} - \frac{1}{3} \delta_{ij}) \delta^3(\mathbf{x}) \supset \propto \frac{1}{x^3}. \quad (II.37)$$

This means that we have nonlocality on scales larger than the nonlocality scale of galaxy IA, which shouldn’t be possible.

II.2.4 E and B modes of spin-2 field

A field of spin-2 tensors on the sphere can also be decomposed into E and B modes. E modes correspond to a divergence like field while B modes create a curl like field. Their key property is that, under a parity transformation, B modes change sign while E modes stay the same.

To calculate these modes, first decompose the spin-2 field into spin-2 spherical harmonics. We are now working on the sphere, so the spin-2 tensors, i.e. the shear, is given by a complex number $_{\pm 2}X(\hat{\mathbf{n}})$. It transforms under a local rotation by an angle α as

$$_{\pm 2}X(\hat{\mathbf{n}}) \rightarrow e^{\mp 2i\psi} _{\pm 2}X(\hat{\mathbf{n}}). \quad (II.38)$$

The decomposition to spin 2 spherical harmonics is given through the usual inner product. We thus obtain spin $a_{\pm 2, lm}$ ’s. We define the E and B modes through their spherical harmonic coefficients as

$$E_{lm} = -\frac{1}{2} (a_{2, lm} + a_{-2, lm}) \quad (II.39)$$

$$B_{lm} = \frac{i}{2} (a_{2, lm} - a_{-2, lm}). \quad (II.40)$$

II.3 EFT of Galaxy IA

II.3.1 Local Deterministic Terms

To get the EFT of galaxy IA, we use $\Pi^{[n]}$ again (equation ...) but instead of making scalar combinations we make symmetric trace free combinations of the form

$$Q_{ij}(\mathbf{x}, \eta) = \sum_O b_O(\eta) O_{ij}(\mathbf{x}, \eta). \quad (\text{II.41})$$

Up to order 3 we get the following local deterministic contributions [5]:

1st order: $\text{tf}[\Pi^{[1]}]_{ij}$

2nd order: $\text{tf}[\Pi^{[2]}]_{ij}, \text{tf}[(\Pi^{[1]})^2]_{ij}, \text{tf}[\Pi^{[1]}]_{ij} \text{tr} \Pi^{[1]}$

3rd order: $\text{tf}[\Pi^{[3]}]_{ij}, \text{tf}[(\Pi^{[1]} \Pi^{[2]})]_{ij}, \text{tf}[\Pi^{[2]}]_{ij} \text{tr} \Pi^{[1]}, \text{tf}[(\Pi^{[1]})^3]_{ij}, \text{tf}[(\Pi^{[1]})^2]_{ij} \text{tr} \Pi^{[1]}, \text{tf}[\Pi^{[1]}]_{ij} (\text{tr} \Pi^{[1]})^2, \text{tf}[\Pi^{[1]}]_{ij} \text{tr}((\Pi^{[1]})^2)$

We do not need to include factors of the form $\text{tr}(\Pi^{[n]})$ for $n > 1$. This was first argued and shown explicitly up to $n = 3$ in [2]. In short, we do not need to include this factor because by using the continuity, Euler, and Poisson equations one can write $\text{tr}(\Pi^{[n]})$ as other terms included elsewhere in the bias expansion.

II.3.2 Non-local deterministic Terms

As explained in subsection ... we should also consider non-local terms, which are suppressed by orders of $(R_* k)^2$ (or $(R_* \nabla)^2$ in real space). At one loop order (when calculating power spectra) it suffices to go up to first order in nonlocality [2], in which case the only new term in our basis is

$$R_*^2 \nabla^2 \text{tf}(\Pi^{[1]})_{ij}.$$

The ratio between non-local and SPT contributions to a power spectrum can be calculated as ...

TODO here: explain this, justify why only one order is needed, give indication to what order of SPT this corresponds

II.3.3 Stochastic Terms

Due to our lack of knowledge about small scale modes, we get stochastic terms in our basis. In analogy to stochastic contributions to galaxy bias [1], we generalize equation II.41 to [5]

$$g_{ij}(\mathbf{x}, \tau) = \sum_O \left[b_O^{(g)}(\tau) + \epsilon_O(\mathbf{x}, \tau) \right] O_{ij}(\mathbf{x}, \tau) + \epsilon_{ij}^O(\mathbf{x}, \tau) O(\mathbf{x}, \tau) + \epsilon_{ij}(\mathbf{x}, \tau). \quad (\text{II.42})$$

The ϵ fields are uncorrelated with the O_{ij} , have vanishing expectation value and are completely described by their one-point distribution, i.e. $\langle \epsilon_O(\mathbf{x}) \epsilon_{O'}(\mathbf{x}') \rangle \propto \delta(\mathbf{x} - \mathbf{x}')$. Each ϵ_{ij} respects the same symmetries as g_{ij} . The O 's are the scalar terms that one would consider in e.g. a galaxy bias EFT. ϵ_{ij} describes the leading order shape noise (when projected to the sky), see also subsection In general, the stochastic contributions up to third order are given by

1st order: ϵ_{ij} ,

2nd order: $\epsilon_{ij}^{\delta} \text{tr} [\Pi^{[1]}], \epsilon_{\Pi^{[1]}} \text{tf} [\pi^{[1]}]_{ij},$

3rd order: $\epsilon_{ij}^{\delta^2} (\text{tr} [\Pi^{[1]}]), \epsilon_{ij}^{K^2} \text{tr} [(\Pi^{[1]})^2], \epsilon_{\Pi^{[2]}} \text{tf} [\Pi^{[2]}]_{ij}, \epsilon_{[\Pi^{[1]}]^2} \text{tf} [(\Pi^{[1]})^2]_{ij},$

$\epsilon_{\Pi^{[1]}\Pi^{[2]}} \text{tf} [\Pi^{[1]}]_{ij} \text{tr} [\Pi^{[1]}].$

Here K_{ij} is the tidal tensor. It is the traceless part of the hessian of the gravitational potential. In comoving coordinates it is defined as

$$K_{ij}(\mathbf{x}, \tau) := \left(\frac{\partial_i \partial_j}{\nabla^2} - \frac{1}{3} \delta_{ij} \right) \delta(\mathbf{x}, \tau). \quad (\text{II.43})$$

II.3.4 Selection Effects

Observed number counts and shapes can also depend on the orientation of the galaxy w.r.t. the line of sight. This is, for example, because galaxies stretched along the line of sight appear more luminous and thus are detected more often than other galaxies at similar distances and of similar intrinsic luminosity. Ref. [6] provides the complete enumeration of these terms for galaxy number density bias, while ref. [5] did the same for galaxy shape density bias. The exact contributions fall out of the scope of this review and can be found in [5]. They all consist of contractions of $\hat{\mathbf{n}}^k \hat{\mathbf{n}}^l$ with $\Pi_{kl}^{[n]}$

TODO here still:

1. also mention renormalization of coefficients

II.3.5 Physical Motivation - Tidal alignment and tidal torquing model

TODO here:

1. Physically motivate linear and quadratic terms by summarizing the argument in [7].

II.3.6 Connection to E and B modes

The 3D galaxy IA field (so not projected) is invariant under parity transformations up to stochastic contributions. This is because $\Pi^{[n]}$ is invariant under parity transformations. When projecting to the sky, one still obtains B -modes (due to the projection, obviously). To calculate the projected power-spectra, start with the powerspectrum of the 3D galaxy IA field, then project it to the sky using eq. One can show (see section 2.6 of [4]) that

$$P_{\delta E}(k, \mu) = \frac{1}{2} \sqrt{\frac{3}{2}} (1 - \mu^2) P_{02}^{(0)}(k), \quad (\text{II.44})$$

$$P_{EE}(k, \mu) = \frac{3}{8} (1 - \mu^2)^2 P_{22}^{(0)}(k) + \frac{1}{2} \mu^2 (1 - \mu^2) P_{22}^{(1)}(k) + \frac{1}{8} (1 + \mu^2)^2 P_{22}^{(2)}(k), \quad (\text{II.45})$$

$$P_{BB}(k, \mu) = \frac{1}{2} (1 - \mu^2) P_{22}^{(1)}(k) + \frac{1}{2} \mu^2 P_{22}^{(2)}(k). \quad (\text{II.46})$$

Here $\mu = \hat{\mathbf{n}} \cdot \hat{\mathbf{n}}$ is the cosine of the angle between $\hat{\mathbf{n}}$ and $\hat{\mathbf{k}}$ and δ is the usual fractional matter density perturbation.

At leading order in the bias expansion (and ignoring stochastic contributions) the B -mode power spectrum vanishes. This is also the case for the LA and NLA models [4]. We thus require an EFT

taken to atleast 2nd order to model B -modes of galaxy IA. As usual, the EB correlator vanishes due to invariance under parity transformations. Parity breaking primordial graviational waves would in general cause a nonzero EB correlator. This would thus serve as a smoking gun for parity violating models of inflation [8].

II.3.7 Results for Three-Dimensional IA Power Spectra

The helicity power spectra can be seperated as

$$P_{\ell\ell'}^{(m)}(k) = [P_{\ell\ell'}^{(m)}]_{\text{L+H.D.}}(k) + [P_{\ell\ell'}^{(m)}]_{(22)}(k) + [P_{\ell\ell'}^{(m)}]_{(13)+(31)}(k) + [P_{\ell\ell'}^{(m)}]_{\epsilon}(k). \quad (\text{II.47})$$

The first term corresponds to leading order and higher derivative contributions (see subsection ...) and the last term corresponds to the noise spectra (see subsection II.2.3). There is no (12) term because this would correspond to contributions like $\langle \delta^3 \rangle$ which vanish or, if we do not assume non-gaussianity, are very small.

II.4 Lensing corrections to observed galaxy density

These are corrections to the g signal without flux/size cutoffs Based on [3]. Unlensed photon geodesic through the centre ($x^\mu = 0$) given by

$$x^\mu(\chi) = (\eta_0 - \chi, \hat{n}\chi), \quad \chi \text{ affine parameter.}$$

Therefore, if we observe a (in general) lensed light ray in direction $\hat{n}^i$ corresponding to an object with redshift $\tilde{z}$ corresponding to the comoving radial distance $\tilde{\chi}(\tilde{z})$, then the observed location of the object is given by $\tilde{x}^\mu$ with

$$\tilde{x}^0 = \eta_0 - \tilde{\chi}, \quad \tilde{x}^i = \hat{n}^i \tilde{\chi}. \quad (\text{II.48})$$

Now define the difference between the observed and the actual object location as the true location minus the observed location, i.e.

$$\Delta x^\mu = x^\mu - \tilde{x}^\mu. \quad (\text{II.49})$$

We also define δz through the inferred emission scale factor $\tilde{a} = 1/(1 + \tilde{z})$ as

$$\frac{a(x^0)}{\tilde{a}} = 1 + \delta z \implies \bar{z} - \tilde{z} = -(1 + \tilde{z})\delta z. \quad (\text{II.50})$$

In other words, δz is the the difference between the observed redshift and the redshift $\bar{z}$ that would be observed in an unperturbed universe.

For the rest of the section, we work in the SC gauge (what is the significance of this? should I say more about it?)

II.4.1 Observed galaxy number density

The observed number of galaxies in a volume $\tilde{V}$ defined in terms of the observed coordinates is given gauge-invariantly as

$$N = \int_{\tilde{V}} J^\mu d\tilde{V}_\mu \quad (\text{II.51})$$

$$= \int_{\tilde{V}} \sqrt{-g} n_g u^\mu \epsilon_{\mu\alpha\beta\gamma} \frac{\partial x^\alpha}{\partial \tilde{x}^1} \frac{\partial x^\beta}{\partial \tilde{x}^2} \frac{\partial x^\gamma}{\partial \tilde{x}^3} d\tilde{x}^1 d\tilde{x}^2 d\tilde{x}^3 \quad (\text{II.52})$$

where $J^\mu = \sqrt{-g}n_g u^\mu$ is the galaxy number 4-current density, $\tilde{V}$ is a hypersurface in 4-space, i.e. a volume in 3-space, and $d\tilde{V}_\mu$ is the normal vector of $\tilde{V}$ with area $d\tilde{V}$. It is mathematically as

$$d\tilde{V}_\mu = \underbrace{\epsilon_{\mu 123} dx^1 dx^2 dx^3}_{\text{in true coordinates}} = \underbrace{\epsilon_{\mu\alpha\beta\gamma} \frac{\partial x^\alpha}{\partial \tilde{x}^1} \frac{\partial x^\beta}{\partial \tilde{x}^2} \frac{\partial x^\gamma}{\partial \tilde{x}^3}}_{\text{in observed coordinates}} d\tilde{x}^1 d\tilde{x}^2 d\tilde{x}^3. \quad (\text{II.53})$$

Finally, n_g is the proper number density of galaxies, i.e. as measured by an observer moving with the galaxies, and u_μ is the 4-velocity of galaxies (we assume zero velocity dispersion) (check this statement).

In the SC gauge, the 4-velocities reduce to $(1/a, 0, 0, 0)$. We thus get

$$N = \int_{\tilde{V}} \sqrt{-g} n_g(x^\alpha) \frac{1}{a(x^0)} \left| \frac{\partial x^i}{\partial \tilde{x}^j} \right| d^3 \tilde{\mathbf{x}}, \quad (\text{SC gauge}). \quad (\text{II.54})$$

One can show that, to first order,

$$\left| \frac{\partial x^i}{\partial \tilde{x}^j} \right| = 1 + \frac{\partial \Delta x^i}{\partial \tilde{x}^i}, \quad \sqrt{-g} = a^4 \left(1 + \frac{1}{2} \delta g^\mu_\mu \right). \quad (\text{II.55})$$

Also,

$$a^3(\tilde{z}) n_g(\mathbf{x}, \tilde{z}) = a^3(\tilde{z}) \bar{n}_g(\tilde{z}) [1 + \delta_g^{sc}(\mathbf{x}, \tilde{z})], \quad (\text{II.56})$$

where we assumed that $\langle \tilde{z} \rangle = \langle \bar{z} \rangle$ and we define δ_g^{sc} as the galaxy number density perturbation in the comoving frame. Additionally, up to first order,

$$a^3(z) n_g(\mathbf{x}, \bar{z}) = a^3(\tilde{z}) \bar{n}_g(\tilde{z}) [1 + \delta_g^{sc}(\mathbf{x}, \tilde{z})] - \underbrace{(1 + \tilde{z}) \frac{d(a^3 \bar{n}_g)}{d\tilde{z}} \Big|_{\tilde{z}=\bar{z}}}_{:=b_c} \delta z \quad (\text{II.57})$$

Now for the important part. The observed galaxy density is defined via the number of galaxies N observed in a volume $\tilde{V}$ as

$$\int_{\tilde{V}} a^3(\tilde{z}) \tilde{n}_g(\tilde{\mathbf{x}}, \tilde{z}) d^3 \tilde{\mathbf{x}} = N. \quad (\text{II.58})$$

Equating the integrand to that of equation II.54 gives

$$a^3(\tilde{z}) \tilde{n}_g(\tilde{\mathbf{x}}, \tilde{z}) = \sqrt{-g} \frac{1}{a(\tilde{z})} n_g(\mathbf{x}, \tilde{z}) \left| \frac{\partial x^i}{\partial \tilde{x}^j} \right| \quad (\text{II.59})$$

using the earlier expansions it can then be shown that we obtain the relation

$$\tilde{\delta}_g = \delta_g^{sc} + b_c \delta z + 2 \frac{\Delta x_{\parallel}}{\tilde{\chi}} + \partial_{\parallel} \Delta x_{\parallel} - 2\hat{\kappa}, \quad \hat{\kappa} := -\frac{1}{2} \partial_{\perp i} \Delta x_{\perp}^i. \quad (\text{II.60})$$

Here “parallel” means parallel to the $\hat{n}$ axis.

Now what is the significance of these corrections? If you take the fourier transform of equation II.60, you get additional terms that look like the terms obtained from nonzero f_{NL} , i.e.

$$\dots \quad (\text{II.61})$$

The effect corresponds to a f_{NL} of order 1. These corrections are thus only relevant for extremely large volume surveys, such as Euclid.

III Modelling Galaxy B-modes

III.1 Point Spread Function (PSF) Effects

III.1.1 Introduction

The PSF describes how points are smeared out due to telescope optics, atmospheric effects, etc. It is modeled as a convolution of the true image. It is typically measured by measuring the shapes of stars. They should appear perfectly spherical (they are located in the Milky Way so haven't been appreciably lensed), meaning that any ellipticity can be attributed to the PSF. The PSF changes over time (e.g. due to changes in the internal temperature of the telescope) and over position in the capture image (typically there is more distortion near the edge of an image). See figure III.1.

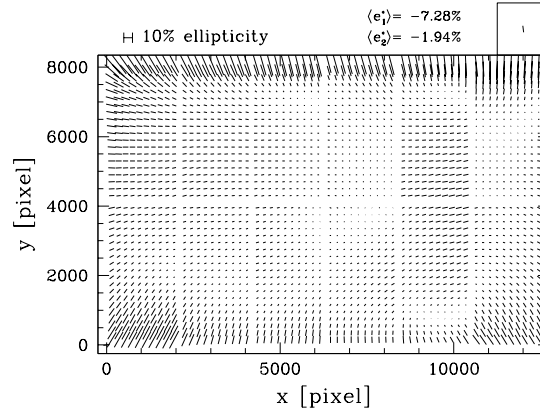


Figure III.1: A visualization of the spatial variation of the PSF ellipticity across the image field, shown as a grid of small vectors indicating the magnitude and orientation of distortions at different pixel positions. Taken from [9].

Therefore, one typically measures the ellipticity of many stars during each exposure and creates a space-variable PSF through some interpolation scheme of the deduced PSF from each star, see e.g. [9] for a discussion.

III.1.2 Change in ellipticity due to PSF

A derivation of how to actually calculate the change in ellipticity, δe , due to arbitrary (but reasonably weak) PSF was first given in ref. [10]. Some errors were then corrected in ref. [11]. We start with a weighted quadrupole,

$$Q_{ij} \equiv \int d^2\theta W(\theta) \theta_i \theta_j f(\vec{\theta}). \quad (\text{III.1})$$

Here $f(\vec{\theta})$ is the flux of the image, $W(\theta)$ is a window function, and we work in coordinates such that the dipole vanishes (center of mass coordinates). The PSF is separated into a spherically symmetric

part and small but purely anisotropic part. Consider the polarization of an image, e_α given by

$$e_1 \equiv \frac{Q_{11} - Q_{22}}{T}, \quad e_2 \equiv \frac{2Q_{12}}{T}, \quad T \equiv Q_{11} + Q_{22}. \quad (\text{III.2})$$

These are invariant to the spherically symmetric part (TODO: check if this is actually true) and so we focus only on the anisotropic part. f is changed due to the PSF as

$$f^{\text{PSF}}(\vec{\theta}) = (PSF * f)(\vec{\theta}) = \int d^2\theta' PSF(\vec{\theta}') f(\vec{\theta} - \vec{\theta}'). \quad (\text{III.3})$$

Plugging this into the quadrupole and Taylor expanding gives

$$\delta Q_{ij} = q_{lm} Z_{lmij}, \quad (\text{III.4})$$

where

$$q_{lm} \equiv \int d^2\theta \theta_l \theta_m PSF(\vec{\theta}), \quad (\text{III.5})$$

$$Z_{lmij} \equiv \int d^2\theta W(\theta) \theta_i \theta_j \partial_l \partial_j f(\vec{\theta}) = \int d^2\theta f(\vec{\theta}) z_{lmij}, \quad (\text{III.6})$$

$$z_{lmij} \equiv \frac{\partial [W(\theta) \theta_i \theta_j]}{\partial_l \partial_m}, \quad (\text{III.7})$$

where the last line is due to integration by parts. The next step would be to calculate δe_α , which can ultimately be written as a function of δQ_{ij} . In the literature a different basis is used, however. Consider that due to the PSF being purely anisotropic q_{ij} must be traceless. Define p_α such that

$$\begin{bmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} p_1 & p_2 \\ p_2 & -p_1 \end{bmatrix}. \quad (\text{III.8})$$

Then one can relate δe_α to p_α through

$$\delta e_\alpha = P_{\alpha\beta}^{\text{PSF}} p_\beta, \quad (\text{III.9})$$

$$P_{\alpha\beta}^{\text{PSF}} \equiv X_{\alpha\beta}^{\text{PSF}} - e_\alpha e_\beta^{\text{PSF}}, \quad (\text{III.10})$$

$$X_{\alpha\beta}^{\text{PSF}} \equiv \frac{1}{T} \int d^2\theta \begin{bmatrix} W + 2W'\theta^2 + W''(\theta_1^2 - \theta_2^2)^2 & 2W''(\theta_1^2 - \theta_2^2)\theta_1\theta_2, \\ 2W''(\theta_1^2 - \theta_2^2)\theta_1\theta_2 & W + 2W'\theta^2 + 4W''\theta_1^2\theta_2^2, \end{bmatrix} f(\vec{\theta}), \quad (\text{III.11})$$

$$e_\alpha^{\text{PSF}} \equiv \frac{1}{I_{11} + I_{22}} \int d^2\theta, \quad (\text{III.12})$$

where prime denotes differentiation with respect to θ^2 . Again, all that was done here is relating δe_α to δQ_{ij} and writing out terms explicitly, simply expressed in a different basis.

If the anisotropic part of the PSF is too large compared to the accuracy desired for a survey, one instead has to work non-perturbatively. See reference [12].

III.1.3 Effects of imperfect PSF compensation

Clearly the PSF will never be perfectly compensated for. Ref.

TODO: relation to B -modes? What happens if we don't compensate for PSF exactly? Check if there is some work on this.

III.2 Masking / window function effects

III.2.1 Modification to shear

Given a set of observed galaxies, the shear field is given by

$$\gamma(\hat{n}) = \sum_{g \in \text{observed gals}} e_g \delta(\hat{n} - \hat{n}_g), \quad (\text{III.13})$$

here e_g is the ellipticity of the galaxy. Now assume the most general version of masking, which is a window function $W(\hat{n})$, i.e. at $\hat{n}$ only a fraction $W(\hat{n})$ of the surveyed galaxies are taken into account¹. The observed shear is then

$$\gamma^{\text{obs}}(\hat{n}) = W(\hat{n})\gamma(\hat{n}). \quad (\text{III.14})$$

In spherical harmonic space this gives

$$\gamma_{\ell m}^{\text{obs}} = \int d^2\Omega {}_2Y_{\ell m}^*(\hat{n})W(\hat{n})\gamma(\hat{n}) \quad (\text{III.15})$$

$$= \int d^2\Omega {}_2Y_{\ell m}^*(\hat{n})W(\hat{n}) \left(\sum_{\ell' m'} {}_2Y_{\ell' m'}(\hat{n})\gamma_{\ell' m'} \right) \quad (\text{III.16})$$

$$= \sum_{\ell' m'} \left(\int d^2\Omega {}_2Y_{\ell m}^*(\hat{n})W(\hat{n}) {}_2Y_{\ell' m'}(\hat{n}) \right) \gamma_{\ell' m'}. \quad (\text{III.17})$$

i.e. just a matrix multiplication of the true multipole moments. ${}_nY_{\ell m}$ is the spin n spherical harmonic.

III.2.2 Effects from uncertainty in the window function

In theory one could simply solve the system of equations shown in equation III.17. In practice the window function is not perfectly known, leading to bias in estimated shear. This is discussed in reference [13], among others. The reasoning proceeds as follows. Assume multiplicative and additive bias to the shear, i.e.

$$\gamma^{\text{obs}}(\hat{n}) = [1 + m(\hat{n})]\gamma(\hat{n}) + c(\hat{n}). \quad (\text{III.18})$$

A spherical harmonic transform as done to obtain equation III.17 gives

$$\gamma_{\ell m}^{E,\text{obs}} = \gamma_{\ell m}^E + \sum_{\ell' m'} [\gamma_{\ell' m'}^E W_{\ell\ell' mm'}^+ + \gamma_{\ell' m'}^B W_{\ell\ell' mm'}^-] + c_{\ell m}^E, \quad (\text{III.19})$$

$$\gamma_{\ell m}^{B,\text{obs}} = \gamma_{\ell m}^B + \sum_{\ell' m'} [\gamma_{\ell' m'}^B W_{\ell\ell' mm'}^+ - \gamma_{\ell' m'}^E W_{\ell\ell' mm'}^-] + c_{\ell m}^B. \quad (\text{III.20})$$

Here

$$W_{\ell\ell' mm'}^+ \equiv \frac{1}{2} [{}_2W_{\ell\ell' mm'} + {}_{-2}W_{\ell\ell' mm'}^*], \quad (\text{III.21})$$

$$W_{\ell\ell' mm'}^- \equiv \frac{i}{2} [{}_2W_{\ell\ell' mm'} - {}_{-2}W_{\ell\ell' mm'}^*], \quad (\text{III.22})$$

and

$${}_sW_{\ell\ell' mm'} \equiv \int d^2\Omega {}_sY_{\ell' m'}^*(\hat{n})m(\hat{n}) {}_sY_{\ell m}(\hat{n}). \quad (\text{III.23})$$

¹Or, alternatively, they are weighed by a factor $W(\hat{n})$.

It is immediately clear that there is mixing between E and B modes, in particular, B modes will be observed even if they are not present in the true shear field. The next question is how the power spectra are changed. We work with the usual definition

$$C_\ell^{XY,\text{obs}} = \frac{1}{2\ell+1} \sum_m \gamma_{\ell m}^{X,\text{obs}} \gamma_{\ell m}^{Y,\text{obs}}, \quad X, Y \in \{E, B\}, \quad (\text{III.24})$$

and assume $C_\ell^{EB} = 0$. This gives a lengthy set of equations, see equation 7 of [13]². If we assume the bias terms are small and work to linear order in them, then the equations simplify considerably.

$$C_\ell^{EE,\text{obs}} = (1 + 2\langle \text{Re}[m(\hat{n})] \rangle) C_\ell^{EE} + 2C_\ell^{c_E E}, \quad (\text{III.25})$$

$$C_\ell^{EB,\text{obs}} = -\langle \text{Im}[m(\hat{n})] \rangle C_\ell^{EE} + \langle \text{Re}[m(\hat{n})] \rangle C_\ell^{BB} + 2C_\ell^{c_E B}, \quad (\text{III.26})$$

$$C_\ell^{BB,\text{obs}} = (1 + 2\langle \text{Re}[m(\hat{n})] \rangle) C_\ell^{BB} + 2C_\ell^{c_B B}. \quad (\text{III.27})$$

Here $C_\ell^{c_X Y}$ is just the power spectrum with $c_{\ell m}^X$ moments instead of $\gamma_{\ell m}^X$.

III.3 B -modes due to intrinsic alignment

III.4 Lensing corrections (post-Born)

III.5 B -modes due to galaxy clustering

From [14]. In the presence of a galaxy correlation function $\delta_D(z_1 - z_2)g(z_1; \theta_1)$ ($\theta = |\vec{\theta}_1 - \vec{\theta}_2|$) we get a modified source redshift distribution of pairs of galaxies at redshifts z_1 and z_2 , i.e.

$$p(z_1, z_2; \theta) = \frac{p(w_1; \theta)p(w_2; \theta)[1 + \delta_D(w_1 - w_2)g(w_1; \theta)]}{1 + \omega(\theta)},$$

where

$$\omega(\theta) = \int_0^\infty dz p(z; \theta)^2 g(z; \theta).$$

Defining ratio of distance to source and distance from redshift z to source as

$$R(z, z_s) = \frac{\chi(z_s) - \chi(z)}{\chi(z_s)},$$

one can obtain the following equation in configuration space (eq 49 of [14])

$$\xi_\pm = \left(\frac{3H_0^2 \Omega_0}{2c^2} \right)^2 \int_0^\infty \frac{dz}{a^2(z)} \int \frac{d\ell}{2\pi} \ell P_\delta \left(\frac{\ell}{\chi(z)}, z \right) J_{0,4}(\ell\theta) \langle R(z, z_1) R(z, z_2) \rangle.$$

The $R(\cdot, \cdot)$ correlation is then modified due to galaxy clustering as

$$\langle R(z, z_1) R(z, z_2) \rangle = \frac{W^2(z) + V(z, \theta)}{1 + \omega(\theta)},$$

²They also include a constant multiplicative bias term, m_0 . i.e. $\gamma^{\text{obs}}(\hat{n}) = [1 + m_0 + m(\hat{n})] \gamma(\hat{n}) + c(\hat{n})$. Clearly our explanation is still completely general, as m_0 could always be absorbed into $m(\hat{n})$.

where

$$W(z) = \int_z^\infty z_1 p(z_1) \frac{\chi(z_1) - \chi(z)}{\chi(z_1)}$$

$$V(w, \theta) = \int_z^\infty dz_1 p^2(z_1) \left(\frac{\chi(z_1) - \chi(z)}{\chi(z_1)} \right)^2 g(z_1; \theta)$$

Clustering thus introduces $V(\cdot, \cdot)$, which leads to B -modes (and corrections to E -modes).

To get the B -modes, first define

$$P_\kappa(\ell) = \left(\frac{3H_0^2 \Omega_0}{2c^2} \right)^2 \int_0^\infty \frac{dz}{a^2(z)} W^2(z) P_\delta \left(\frac{\ell}{\chi(z)}, z \right),$$

$$P_c(\ell, \theta) = \left(\frac{3H_0^2 \Omega_0}{2c^2} \right)^2 \int_0^\infty \frac{dz}{a^2(z)} V(z, \theta) P_\delta \left(\frac{\ell}{\chi(z)}, z \right).$$

Then define

$$\mathcal{P}(\ell, \theta) = \frac{P_c(\ell, \theta) - w(\theta) P_\kappa(\ell)}{1 + \omega(\theta)}.$$

One then obtains (eq 60 from [14])

$$P_B(\ell) = \frac{1}{2} \int d\theta \theta \int d\ell' \ell' \mathcal{P}(\ell'; \theta) [J_0(\ell\theta) J_0(\ell'\theta) \pm J_4(\ell\theta) J_4(\ell'\theta)].$$

III.6 B -modes due to flux / size cutoffs and lensing

(TODO: Check [15], also check Schmidt et al. 2008; Hui et al. 2007 as cited in [16] for effects on bispectrum/3-point.) Galaxies are selected based on brightness and size cutoffs. This leads to magnification bias and size bias. This gives B -mode correction. See eq. 60 of [16]. On scales $\ell > 50$ lensing bias effectively increases the B -mode power spectrum by approximately a factor $(1 + 2q)$, where q is determined by the slope of the luminosity and radius distribution functions of the galaxy sample.

If we work with a flux and size cutoff and define [17]

$$q = -2 \frac{\partial \ln n_{\text{obs}}}{\partial \ln f} \Big|_{r_{\text{min}}, f_{\text{min}}} - \frac{\partial \ln n_{\text{obs}}}{\partial \ln r} \Big|_{r_{\text{min}}, f_{\text{min}}} - 2,$$

then (at 2nd order in lensing)

$$1 + \delta_{\text{lens}} = \frac{1 + q\kappa^2 + C_1\kappa^2 + C_2|\gamma|^2}{1 + C_1\langle\kappa^2\rangle + C_2\langle|\gamma|^2\rangle}.$$

The measured reduced shear g_{obs} is given in terms of the actual reduced shear g , through $g_{\text{obs}} = g(1 + \delta_{\text{lens}})$. This corrected value is then correlated to obtain B -modes.

IV Calibration and Validation with Simulations

TODO here:

1. Explain validity of EFT of IA based on [\[4\]](#)
2. Take a look at hydrodynamic simulations [\[18, 19\]](#)

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A Simulating/modeling nonlinear structure formation and baryonic feedback with HMCODE

A.1 The halo model

In order to accurately model the matter power spectrum at nonlinear scales, the halo model has been proposed since the early 2000's (TODO: ESSENTIAL REFERENCES HERE). One assumes that all matter is located in exactly one halo. First rewrite the matter density field in the dimensionless quantity Δ^2 :

$$\Delta^2(k) = 4\pi V \left(\frac{k}{2\pi} \right) P(k), \quad (\text{A.1})$$

which gives the fractional contribution to the variance per logarithmic interval in k . In the halo model, the power spectrum can be split into correlations between points within the same halo and between points within different halos. This gives the 1 and 2 halo terms, respectively.

$$\Delta^2(k) = \Delta_{2H}^2 + \Delta_{1H}^2. \quad (\text{A.2})$$

The 1 halo term is given as

$$\Delta_{1H}^2(k) = 4\pi \left(\frac{k}{2\pi} \right)^3 \frac{1}{\bar{\rho}^2} \int_0^\infty M^2 W^2(k, M) F(M) dM. \quad (\text{A.3})$$

The following definitions are in order:

1. $F(M)$ is the halo mass function, it is defined so that $F(M)dM$ is the comoving number density of halos with mass between M and $M + dM$.
2. $W(k, M)$ is the normalized Fourier transform of the halo density profile, $\rho_{halo}(r, M)$, which is assumed to be spherically symmetric:

$$W(k, M) = \mathcal{F} \left[\frac{1}{M} \rho_{halo}(r, M) \right](k) = \frac{1}{M} \int_0^{r_v} \frac{\sin(kr)}{kr} 4\pi r^2 \rho_{halo}(r, M) dr. \quad (\text{A.4})$$

3. r_v is the virial radius, defined such that the average density within a distance r_v from the center of the halo is $\Delta_v \bar{\rho}$, where Δ_v is chosen by the author and is usually around 200.

TODO: 2 halo spectrum def?

To get the halo model to work we then need to choose an accurate halo density profile. A common one is the NFW profile [20],

$$\rho_{NFW}(r) = \frac{\rho_N}{(r/r_s)(1 + r/r_s)^2}, \quad (\text{A.5})$$

where r_s is the scale that roughly separates the core of the halo from the outer portion and ρ_N is a normalization constant. The other choice that needs to be made is for the halo number density. It has been shown to have a near univereal form (TODO: SOME REFS). First define

$$v \equiv \frac{\delta_c}{\sigma(M)}, \quad (\text{A.6})$$

where $\sigma(M)$ is the variance of the mass density smoothed on the comoving scale r_v corresponding to a halo of mass M and δ_c at some redshift z is defined through the linear growth function and deviation function σ .

Now define $f(v)$ so that it relates to $F(v)$ as

$$\frac{M}{\bar{\rho}} F(M) dM = f(v) dv. \quad (\text{A.7})$$

f then has a nearly universal form which was first derived in [21] and improved upon in [22] to better fit simulations. It is given as

$$f(v) = A_f \left[1 + \frac{1}{(av^2)^p} \right] e^{-av^2/2}, \quad a = 0.707, \quad p = 0.3. \quad (\text{A.8})$$

It is accurate up to about 20 percent for a wide range of cosmologies.

A.2 Baryonic feedback

Baryonic feedback refers to effects unique to baryons due to their collisional and warm nature. Important effects include cooling, supernova explosions, and active galactic nuclei (AGN). (TODO: ADD BASIC REFS)

1. Baryonic matter has inelastic collisions. Kinetic energy is lost and converted mainly into heat, which is then converted into radiative energy. This causes baryonic matter to clump up more than CDM, in turn leading to the creation of macroscopic structures such as stars and planets, as well as to denser halo cores than expected from a CDM only universe.
2. Active galactic nuclei means a galaxy with an active ("feeding") supermassive black hole at its center. Matter is sucked into accretion disks, where it is heated up. This releases a large amount of energy in the form of radiation and jets (relativistic plasma) back into the halo, thereby expanding ("bloating") it. AGN feedback can reduce the baryon fraction in the centres of haloes by a factor of 2 in the most extreme models [23].
3. Supernova explosions have essentially the same effect on halo structure. Energy is released into the halo in the form of radiation, heating up the halo and bloating it.

Constraints from the Dark Energy Survey could be strongly biased if the most extreme feedback scenarios apply and constraints from Euclid would be severely biased for any feasible feedback scenario. The halo model is particularly good at taking into account baryonic feedback, as its main effect is to change halo shape, which we can easily change when using this model, [23, 24, 25].

A.3 Optimizing the halo model and taking into account baryonic feedback

The halo model has been significantly improved upon by adding degrees of freedom and fitting to simulations in [26] and later with improved accuracy in [27]. In particular, they take into account BAO damping, massive neutrinos and baryonic feedback. The key changes made are:

1. For the 2 halo term, they use "de-wiggled" linear theory. The BAO "wiggles" are isolated from

the rest of the power spectrum signal,

$$P^{lin}(k, z) = P_{smt}(k, z) + P_{wig}(k, z) \quad (A.9)$$

then the de-wiggled linear power spectrum has the BAO wiggles damped at small scales by removing part of that contribution as follows:

$$P_{dwl}(k, z) = P^{lin}(k, z) - (1 - e^{-k^2 \sigma_v(z)}) P_{wig}(k, z). \quad (A.10)$$

This damping models the BAO damping that occurs due to non-linear evolution. 2 halo term is then given in terms of this de-wiggled power spectrum. The full 2 halo equation (TODO: REF THE EQ ONCE I WRITE IT DOWN) would likely be more accurate but is significantly more costly to compute due to it containing an integral.

2. In order to obtain an accurate approximation of the 2 halo term, further damping of the power spectrum as a whole is required on smaller scales. Nonlinear collapse transfers power from large and intermediate scales into very small scales, causing a slight suppression at $k \sim 0.03$, $z = 0$ and an enhancement at high k . This suppression is modeled as

$$P^{2H}(k, z) = P_{dwl}(k, z) \left[1 - f \frac{(k/k_d)^{n_d}}{1 + (k/k_d)^{n_d}} \right], \quad (A.11)$$

where f , n_d , and k_d are free parameters.

3. On large scales the 1 halo term is too large and actually outgrows the linear / 2 halo term. This is because halos are assumed to be randomly distributed over space, which leads to a poisson noise contribution. In reality, they are much more uniformly distributed and are not allowed to overlap with one another due to how the halo model is constructed. Damping factor at large scales is thus required to compensate for this deficiency. This is done as

$$P^{1H}(k, z) \rightarrow P^{1H}(k, z) \frac{(k/k_*)^4}{1 + (k/k_*)^4}, \quad (A.12)$$

where k_* is another free parameter.

4. To take into account halo bloating, e.g. from baryonic feedback, the halo mass distribution is modified in a mass dependent way as

$$\rho(r, M) \propto \frac{1}{r / (v^\eta r_s) [1 + r / (v^\eta r_s)]^2}. \quad (A.13)$$

η is now another free parameter. For $\eta > 0$ this modification bloats heavy halos and concentrates light halos. Mathematically, the change corresponds to $W(k, M) \rightarrow W(v^\eta k, z)$.

(Next are some changes made specifically to take into account baryonic feedback.)

5. A delta function of magnitude f_* is placed at the center of halos to model the presence of mass in the form of stars near its center. This was shown to be necessary to accurately calculate the MPS in multiple works.
6. The ratio of virial radius, r_v , and r_s is defined as the function $c \equiv r_v / r_s$. By choosing c we thus choose r_s . c is chosen in line with [28], which found

$$c(M, z) \approx B \left[\frac{1 + z_f(M, z)}{1 + z} \right], \quad (A.14)$$

where z_f is the “formation redshift” related to the linear growth function and matter field variance at relevant scales. B is left as a free parameter (instead of the default $B = 4$) to model gas expulsion lowering halo mass concentration.

7. Gas expulsion is further taken into account by lowering the gas content of halos via

$$f_g(M) = \left(\frac{\Omega_b}{\Omega_m} - f_* \right) \frac{(M/M_b)^\beta}{1 + (M/M_b)^\beta}, \quad (\text{A.15})$$

where f_g is the halo gas fraction, $\frac{\Omega_b}{\Omega_m} - f_*$ is the available gas reservoir, and $M_b, \beta > 0$ are fitted parameters. Massive halos are unaffected while lighter halos are modeled to have lost significant amounts of gas. The NFW window function is then replaced by

$$\tilde{W}(M, k) = \left[\frac{\Omega_c}{\omega_m} + f_g(M) \right] W(M, k) + f_* \frac{M}{\bar{\rho}}. \quad (\text{A.16})$$

In [27] it was observed that when fitting the power spectrum to the BAHAMAS simulations that all 6 baryonic feedback parameters have a roughly linear relationship to the “feedback temperature”, $\log_{10}(T_{AGN}/K)$. This is the temperature at which surrounding gas is heated by AGN. This is thus a much simpler model with only slightly less accuracy (couple of percent accuracy compared to < 2.5% accuracy for the 6 parameter model).